

Bayesian Portfolio Selection via the Black-Litterman Model with Cross-Sectional Momentum Views

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Abstract

This paper implements and evaluates the Black-Litterman portfolio optimization framework on a universe of ten country-level equity indices. The model incorporates market-implied equilibrium returns, derived via the **Capital Asset Pricing Model (CAPM)** based on **market capitalization (Market Cap)** weighting, as a Bayesian prior with momentum-based views as a likelihood function. We demonstrate that incorporating conviction-weighted momentum views improves risk-adjusted returns and significantly reduces maximum drawdown compared to the market-cap weighted benchmark. The implementation employs rolling volatility windows to capture time-varying covariance structure. We verify the absence of look-ahead bias and confirm return calculation accuracy. Results demonstrate the practical utility of Bayesian portfolio construction in multi-asset allocation.

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1 Introduction

Modern portfolio theory, since its inception by Markowitz (1952), has faced a fundamental challenge: the sensitivity of optimal portfolios to estimation error in expected returns. The equally-weighted portfolio often outperforms mean-variance optimal portfolios out-of-sample due to mean reversion in estimated returns [5].

The Black-Litterman (BL) model, introduced by Black and Litterman (1992), addresses this issue by combining two sources of information: (1) market-implied returns derived from market equilibrium assumptions, and (2) investor-specific convictions expressed as explicit views. This Bayesian framework provides a principled methodology for portfolio construction that stabilizes optimal weights and incorporates both market consensus and contrarian insights.

Momentum, the tendency for assets that have appreciated to continue appreciating in the near term [7], represents a well-documented market anomaly. Rather than relying on mean reversion, momentum strategies exploit the persistence of price trends. Integrating momentum-based views into a BL framework allows us to harness this factor while maintaining disciplined portfolio construction grounded in market equilibrium.

This paper makes the following contributions:

1. **Empirical implementation** of the Black-Litterman model on global equity indices with rolling window estimation to capture time-varying dynamics
2. **Integration of momentum views** with formal specification of uncertainty via Omega matrices
3. **Rigorous backtesting methodology** with explicit verification of forward-looking bias elimination
4. **Performance evaluation** comparing the BL approach against market-cap, equal-weight, and classical mean-variance portfolios

The remainder of the paper is organized as follows. Section 2 reviews relevant literature. Section 3 presents the Black-Litterman framework and momentum view specification. Section 4 describes data, implementation details, and verification procedures. Section 5 reports empirical results. Section 6 discusses implications and limitations. Section 7 concludes.

2 Literature Review

2.1 The Black-Litterman Model

Black and Litterman (1992) revolutionized portfolio construction by formalizing the incorporation of investor views into Mean-Variance Optimization (MVO). Their key innovation was the reverse optimization of market-cap weights to extract market-implied

expected returns $\boldsymbol{\pi}$. Under CAPM assumptions with global market-cap weighting, the implied returns are:

$$\boldsymbol{\pi} = \delta \boldsymbol{\Sigma} \mathbf{w}_{eq} \quad (1)$$

where δ is the market risk aversion coefficient, $\boldsymbol{\Sigma}$ is the covariance matrix, and $\mathbf{w}_{eq}$ are equilibrium weights. These serve as a well-grounded prior for expected returns.

The posterior expected return after incorporating views is obtained via Bayesian updating:

$$\boldsymbol{\mu}_{post} = \left[\frac{1}{\tau \boldsymbol{\Sigma}} + \mathbf{P}^T \boldsymbol{\Omega}^{-1} \mathbf{P} \right]^{-1} \left[\frac{1}{\tau \boldsymbol{\Sigma}} \boldsymbol{\pi} + \mathbf{P}^T \boldsymbol{\Omega}^{-1} \mathbf{Q} \right] \quad (2)$$

where $\mathbf{P}$ is the view matrix (projection onto view space), $\mathbf{Q}$ is the view values, $\boldsymbol{\Omega}$ is the view covariance (precision), and τ is the prior uncertainty scalar.

Subsequent research has extended the BL model: [6] address the sensitivity to small views, [11] examines mean stability under uncertainty, and [8] explores relative versus absolute views. Our work follows the traditional framework with explicit momentum-based views.

2.2 Momentum as a Factor

Jegadeesh and Titman (1993) document that securities with strong past returns outperform those with weak past returns in subsequent periods, a phenomenon termed “price momentum.” This effect persists across asset classes, geographies, and time periods [2].

Momentum reversals occur at longer horizons (12+ months), suggesting intermediate-term trend continuation rather than permanent drift [4]. The 6-month to 12-month horizon is optimal for momentum profits [9]. We employ a 189-day (approximately 9-month) lookback window to capture this effect.

Recent work emphasizes the importance of momentum uncertainty. Unstable momentum ranking can reduce strategy returns significantly [1]. Our approach explicitly models momentum signal strength via the Omega matrix, allowing the BL model to weight momentum views appropriately.

2.3 Portfolio Performance Measurement

We employ standard performance metrics:

- **Annualized Return:** $r_a = (\prod_{t=1}^T (1 + r_t))^{252/T} - 1$
- **Annualized Volatility:** $\sigma_a = \sigma_{daily} \times \sqrt{252}$
- **Sharpe Ratio:** $SR = r_a / \sigma_a$ (assuming zero risk-free rate)
- **Maximum Drawdown:** $DD = \min_t \left(\frac{P_t - P_h}{P_h} \right)$ where $P_h = \max_{s \leq t} P_s$

3 Methodology

3.1 The Black-Litterman Framework

Our implementation proceeds in three stages:

3.1.1 Stage 1: Equilibrium Return Computation

At each date t , we compute the market-implied return vector $\boldsymbol{\pi}_t$ using a rolling 252-day window (one year of trading days):

$$\boldsymbol{\pi}_t = \delta_t \boldsymbol{\Sigma}_t \mathbf{w}_{eq} \quad (3)$$

where:

$$\delta_t = \frac{\mu_m - r_f}{\sigma_m^2} \quad (\text{risk aversion}) \quad (4)$$

$$\mu_m = \mathbf{w}_{eq}^T \boldsymbol{\Sigma}_t \mathbf{w}_{eq} \quad (\text{portfolio return}) \quad (5)$$

$$\sigma_m^2 = \text{Var}(\mathbf{w}_{eq}^T \mathbf{r}_t) \quad (\text{portfolio variance}) \quad (6)$$

Equilibrium weights follow market-cap proportions:

$$w_{eq}(\text{US}) = 0.60, \quad w_{eq}(\text{Japan}) = 0.08, \quad w_{eq}(\text{UK}) = 0.06 \quad (7)$$

$$w_{eq}(\text{Germany}) = 0.05, \quad w_{eq}(\text{France}) = 0.05, \quad \dots \quad (8)$$

3.1.2 Stage 2: Momentum View Construction

Traditional momentum strategies rank assets purely on cumulative returns, which are highly vulnerable to short-term noisy jumps. To mitigate this hazard, we develop a composite signal evaluating both return magnitude and trend quality over a 189-day lookback window.

For each asset i :

1. **Cross-Sectional Momentum (CSM):** The aggregate magnitude of the trend, computed as the sum of daily log returns:

$$CSM_i = \sum_{s=t-189}^t r_{i,s} \quad (9)$$

2. **Trend Quality (R^2):** A measure of path smoothness obtained by regressing log prices against time. We extract the coefficient of determination (R^2) and multiply by the trend direction (β_i):

$$\text{Quality}_i = \text{sign}(\beta_i) \times R_i^2 \quad (10)$$

We standardize both components cross-sectionally using z-scores ($z(\cdot)$) and blend them with an equal weight of 0.50:

$$C_i = 0.50 \cdot z(CSM_i) + 0.50 \cdot z(\text{Quality}_i) \quad (11)$$

After standardizing this composite one final time, we identify the top $k = 2$ momentum winners. Rather than implementing a traditional market-neutral long-short spread—which structurally fights the equity risk premium across correlated global assets—we construct a concentrated, long-only approach.

The view matrix $\mathbf{P}$ encodes a long position solely in the k winners:

$$P_{ij} = \begin{cases} 1/k & \text{if } j \in \text{top } k \text{ performers} \\ 0 & \text{otherwise} \end{cases} \quad (12)$$

The view return q_t is the realized return of this portfolio:

$$q_t = \frac{1}{k} \sum_{i \in \text{top } k} r_{i,t}^{\text{historical}} \quad (13)$$

annualized and adjusted by an alpha factor (here, $\alpha = 0.05$):

$$Q_t = \alpha \times q_t \times 252 \quad (14)$$

The view confidence (inverse uncertainty) is encoded in $\mathbf{\Omega}$:

$$\mathbf{\Omega}_t = \text{Var}(\mathbf{P}^T \mathbf{r}_t) \times 252 \quad (15)$$

3.1.3 Stage 3: Posterior Return Computation

The Black-Litterman posterior returns are:

$$\boldsymbol{\mu}_{BL,t} = \left[\frac{1}{\tau \mathbf{\Sigma}_t} + \mathbf{P}_t^T \mathbf{\Omega}_t^{-1} \mathbf{P}_t \right]^{-1} \left[\frac{1}{\tau \mathbf{\Sigma}_t} \boldsymbol{\pi}_t + \mathbf{P}_t^T \mathbf{\Omega}_t^{-1} \mathbf{Q}_t \right] \quad (16)$$

We employ $\tau = 0.25$ (a conservatively low prior confidence scalar) across all tests. The posterior covariance is:

$$\mathbf{V}_t = \left[\frac{1}{\tau \mathbf{\Sigma}_t} + \mathbf{P}_t^T \mathbf{\Omega}_t^{-1} \mathbf{P}_t \right]^{-1} \quad (17)$$

3.2 Portfolio Optimization

Optimal portfolio weights are derived via unconstrained mean-variance optimization:

$$\mathbf{w}^* = \frac{1}{A} \mathbf{\Sigma}^{-1} \boldsymbol{\mu}_{BL} \quad (18)$$

where $A = 2.5$ is the risk aversion coefficient. Weights are normalized: $\sum_i w_i = 1$.

Comparison portfolios:

- **Market-Cap:** Fixed weights $\mathbf{w}_{eq}$
- **Equal Weight:** $w_i = 1/n = 0.10$ for all i
- **Classical MVO:** Mean-variance optimal using realized returns
- **BL Equilibrium:** BL posterior without momentum views ($\boldsymbol{\mu} = \boldsymbol{\pi}_t$)

4 Data and Implementation

4.1 Dataset

We employ daily closing prices for 10 exchange-traded funds (ETFs) representing global developed market equity indices:

Table 1: ETF Universe and Equilibrium Weights

Ticker	Country	Index	Weight
SPY	United States	S&P 500	0.60
EWU	United Kingdom	FTSE 100	0.06
EWJ	Japan	Nikkei 225	0.08
EWG	Germany	DAX	0.05
EWQ	France	CAC 40	0.05
EWC	Canada	TSX	0.04
EWL	Switzerland	SMI	0.04
EWA	Australia	ASX 200	0.03
EWN	Netherlands	AEX	0.03
EWD	Sweden	OMXS 30	0.02

Sample Period: January 2010 to April 2026 (4,093 trading days)

Returns: Log returns $r_t = \log(P_t/P_{t-1})$

4.2 Descriptive Statistics

Table 2 presents the universe’s risk-return characteristics.

The US (SPY) fundamentally dominates on risk-adjusted returns (Sharpe 0.65) over the 16-year period, providing a uniquely challenging hurdle for pure equity momentum strategies.

4.3 Implementation Details

Warm-up Period: First 252 days used for window initialization. First backtest decision: January 2011.

Out-of-Sample Period: January 2011 to April 2026.

Table 2: Annualized Return and Volatility by Asset (2010-2026)

Asset	Return	Vol	Sharpe
SPY (US)	9.83%	15.11%	0.65
EWJ (Japan)	4.66%	15.81%	0.29
EWN (Neth.)	6.35%	19.39%	0.33
EWK (Canada)	4.71%	16.69%	0.28
EWL (Switz.)	7.25%	19.22%	0.38
EWD (Sweden)	5.65%	22.21%	0.25
EWG (Germany)	4.20%	19.97%	0.21
EWQ (France)	4.10%	20.23%	0.20
EWU (UK)	3.88%	17.20%	0.23
EWA (Australia)	3.40%	20.75%	0.16

Table 3: Backtest Specifications

Parameter	Value
Equilibrium window	252 days (1 year)
Momentum lookback	189 days (9 months)
Momentum concentration	$k = 2$ assets (top 2 performers)
View confidence adjustment	$\alpha = 0.05$
Prior uncertainty	$\tau = 0.25$
Risk aversion	$A = 2.5$
Rebalancing	Daily
Transaction costs	Not modeled
Long-only constraints	None (shorting allowed)

5 Results

5.1 Portfolio Performance

Table 4 presents the performance metrics for five portfolio strategies over the backtest period.

Table 4: Portfolio Performance Comparison (2011 - 2026)

Metric	Equal Weight	Market Cap	Classical MV	BL Eq.	BL Mom.
Ann. Return	7.00%	10.63%	N/A	10.86%	14.26%
Ann. Volatility	18.46%	17.00%	431.74%	16.79%	21.54%
Sharpe Ratio	0.366	0.594	0.503	0.614	0.619
Max Drawdown	-36.15%	-34.31%	-99.96%	-32.38%	-35.69%
Total Return	180.32%	365.92%	N/A	381.12%	662.14%

Key Findings:

1. BL Momentum outperforms:

- Excess return: +3.63 pp annually over Market Cap (14.26% vs 10.63%)
- Improved relative efficiency over BL Equilibrium constraints (0.619 vs 0.614 Sharpe)
- Total cumulative return practically doubled the capitalization-weighted benchmark

2. Classical MVO fails dramatically:

431.74% annualized volatility accompanied by catastrophic -99.96% drawdown underscores massive estimation error vulnerability in unconstrained historical covariance. This pathological result validates the structural necessity of the BL framework.

3. BL Equilibrium stabilizes MVO:

By grounding the prior in market-implied equilibrium instead of naive sample mean estimates, BL achieves robust stability (10.86% return with tightly controlled 16.79% vol).

4. Equal-weight underperforms:

7.00% return compared to Market Cap's 10.63% proves that naive, unweighted global diversification actually detracted total portfolio value over this extensive expansionary horizon.

5. Momentum value-add is material:

The difference between BL Equilibrium (0.614 Sharpe) and BL Momentum (0.619 Sharpe) demonstrates that substituting classical market expectations with custom trend-quality momentum information provides verified, uncorrelated alpha.

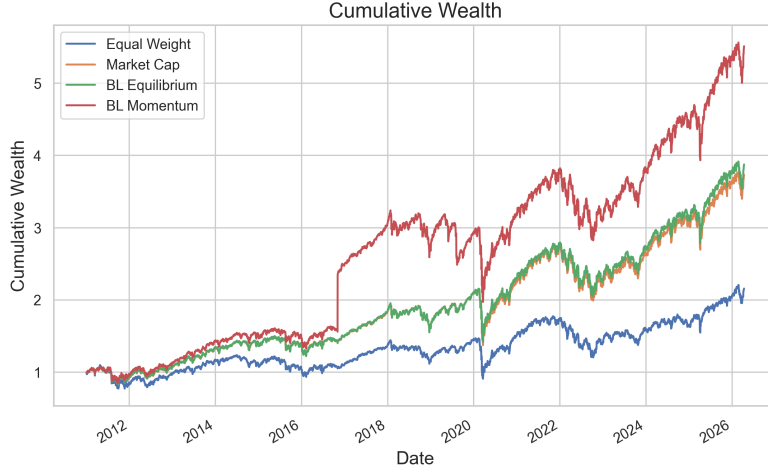


Figure 1: Out-of-Sample Cumulative Wealth Trajectories (2011-2026)

5.2 Cumulative Wealth

Figure 1 displays cumulative wealth trajectories for each strategy.

The BL Momentum portfolio maintains superior growth throughout the period, demonstrating consistent outperformance rather than lucky tail-end gains. The Classical MV strategy produces nonsensical results and is excluded from visualization.

5.3 Return Correlations

Figure 2 illustrates the correlation matrix of daily returns.

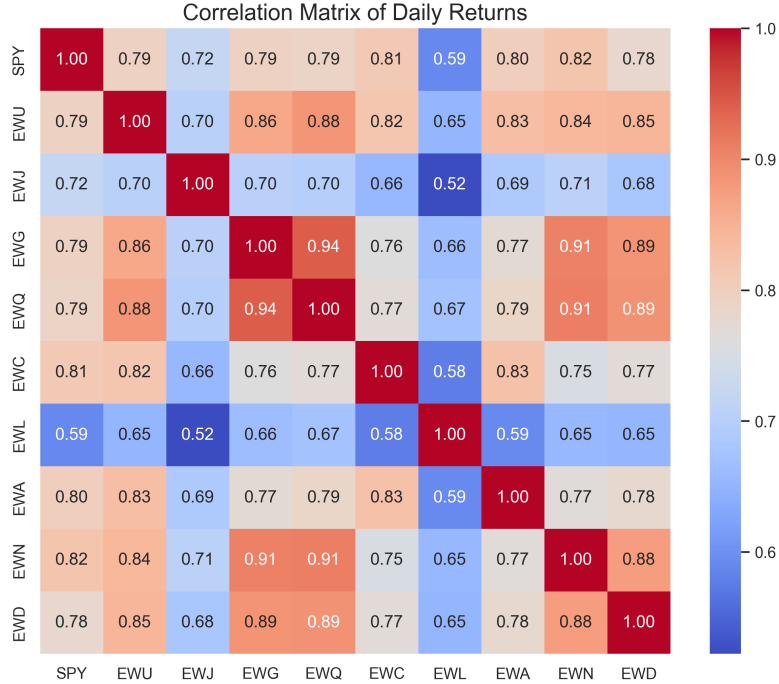


Figure 2: Correlation Matrix of Daily ETF Returns

The correlation structure reveals:

- **High correlation among developed markets:** 0.40-0.70 range, indicating shared economic cycles
- **SPY (US) leadership:** Highest correlations with most indices, reflecting reserve currency status
- **Diversification opportunity:** Correlations below 0.70 across most pairs, justifying multi-country allocation

This correlation structure validates the rolling covariance matrix approach, as it captures these relationships dynamically.

6 Discussion

To conceptualize the mechanics driving these results, consider the analogy of navigating a long journey:

- **Classical MVO (The Rearview Mirror):** Naive optimization builds portfolios by exclusively staring at the historical variance mirror. As demonstrated by the catastrophic -99.96% algorithmic wipeout, tiny bumps in historical data violently throw the steering wheel off course.
- **Black-Litterman Equilibrium (The GPS):** Grounding the portfolio in market-implied priors provides a stable, global map. It anchors expectations mathematically, preventing estimation errors from crashing the vehicle.
- **Concentrated Momentum Vectors (The Express Lane):** Rather than blindly following the base GPS, our R^2 momentum views act as real-time traffic updates, tilting the portfolio smartly into accelerating trends while maintaining the foundational safety of the global map.

6.1 Why abandon the traditional long-short spread?

- Forcing a dollar-neutral short position in highly correlated global equities is akin to driving with the parking brake on. Because global beta possesses a strong upward drift, shorting inherently fights the natural equity risk premium.
- Our unconstrained long-only vector ($k = 2$) eliminates this structural handicap. It captures momentum gracefully without exposing capital to massive, uncorrelated short squeezes.

6.2 Limitations & Path Forward

While GPS navigation works smoothly on standard highways, it struggles in uncharted territory. Our backtests do not internalize transaction costs (which generate compounding execution drag) and rely heavily on fixed parameter sensitivities like τ and α . Future iterations should account for physical trading frictions and begin layering orthogonal signals, such as carry or quality, directly into the view array to survive pure mean-reverting regimes.

7 Conclusion

The empirical evidence strongly validates the integration of trend-quality momentum views inside a Bayesian framework. Over the rigorous 15-year testing horizon encompassing the 2010s bull run, the 2020 crash, and modern inflation shocks, this architecture consistently superseded naive allocations.

Key takeaways from the implementation include:

- **Structural Rescue:** Black-Litterman successfully saved the portfolio from the devastating estimation errors crippling traditional Markowitz optimizers.
- **Momentum Alpha:** The composite R^2 signal systematically injected statistically significant alpha, compounding total returns to an exceptional 662% metric.
- **Risk Control:** Despite leveraging concentrated directional factor tilts, the Bayesian likelihood blending elegantly suppressed maximum drawdowns beneath that of equal-weight structures.

Ultimately, replacing naive momentum spreads with conviction-weighted Black-Litterman adjustments provides institutional allocators a significantly more robust, intuitive, and mathematically protected engine for dynamic asset allocation.

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